1) a) $t_0 = 0$, $y_0 = 100$, $V_0 = 98$

$V = 98 - 9,8t$ $\quad$ $y = 100 + 98t - \frac{1}{2} 9,8t^2$

$0 = 98 - 9,8t \Big\} \rightarrow t = \frac{98}{9,8} = 10s$

$y = 100 + 98 + 10 - \frac{1}{2} 9,8 \cdot 10^2 = 590$

$\boxed{h = 590\,m}$

b) $e_0 = 100$, $V_0 = 98$

$100 = 100 + 98t - \frac{1}{2} 9,8t^2 \rightarrow t(98 - 4,9t) = 0$

$\boxed{t = 20s}$

c) $-100 = 100 + 98t - \frac{1}{2} 9,8t^2 \Big\}$

$\rightarrow t = \frac{98 \pm \sqrt{98^2 + 4 \cdot 4,9 \cdot (-100)}}{9,8}$ $\quad$ $\boxed{t = 20,97s}$ $\rightarrow$ d

$V = 98 - 9,8 \cdot 20,97$

$\boxed{V = -107,53\,m/s}$